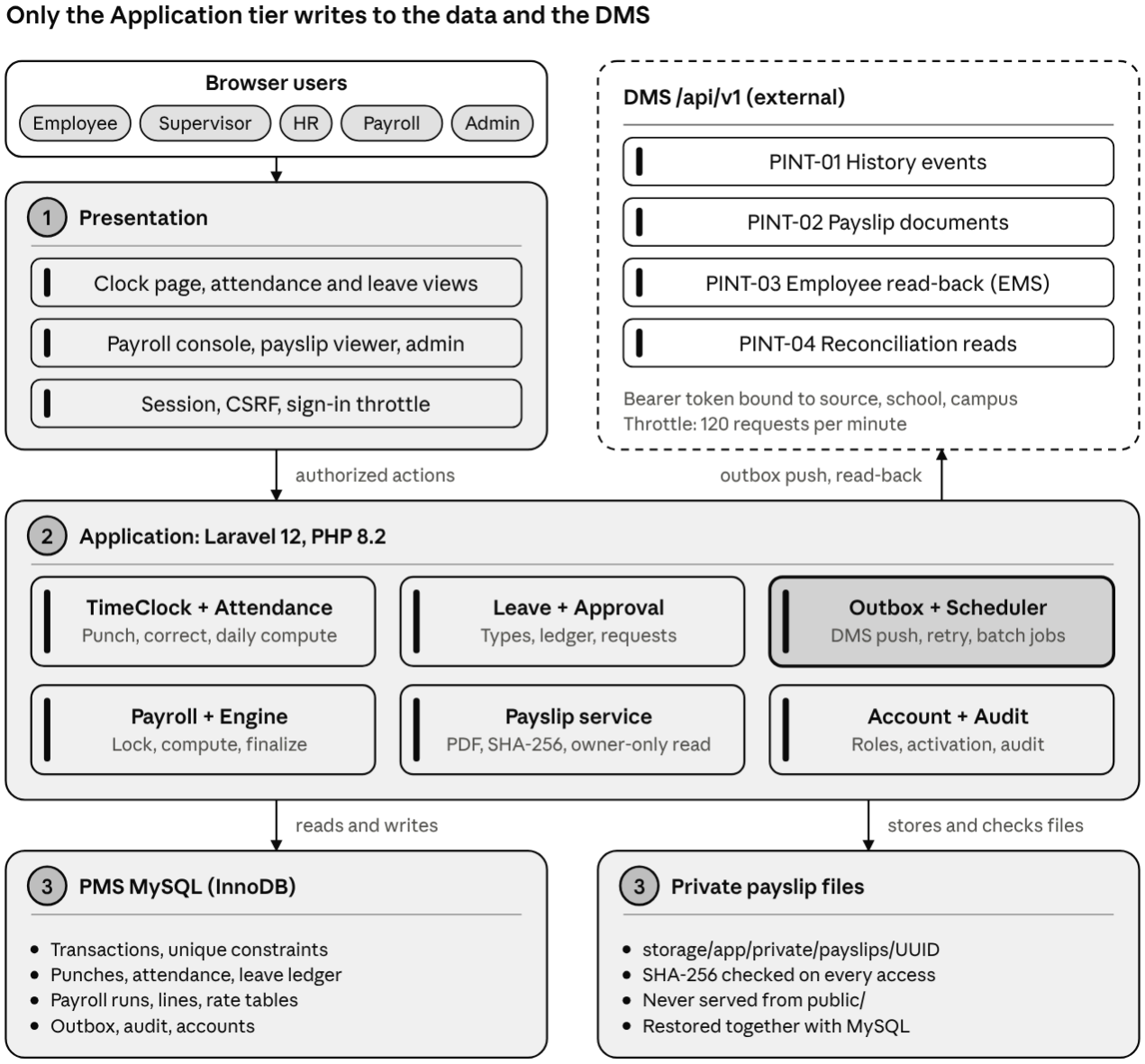
Solution Design Document - Payroll Management System (PMS)

Oct 16, 2026 · @jp

The PMS is one Laravel 12 application split into three tiers (Presentation, Application, Data) that records time in/out, tracks attendance, handles leave requests and generates payslips, then sends payslips and payroll history to the DMS through the /api/v1 boundary.

At a glance



PMS three-tier architecture · 3 tiers, DMS link

Browser users reach the Presentation tier, and only the Application tier writes to MySQL, to the payslip files and to the DMS /api/v1. Presentation never touches the data, and the DMS never reaches into the PMS database.

Scope and objectives

The PMS owns time records, leave and payroll computation. The EMS stays the owner of employee master data, and the DMS keeps the documents and history copies the PMS sends it.

ID	Objective	Measure	Target
OBJ-01	Trustworthy time and attendance records	Punch records changed or deleted after capture	0
OBJ-02	Accurate, repeatable payroll	Locked-input payroll reruns that give a different net pay	0
OBJ-03	Payslips preserved in one controlled place	Payslips of finalized runs also held in the DMS	100%
OBJ-04	Safe retries toward the DMS	Duplicate history records created by a resent event	0
OBJ-05	Low operating cost	Database engines needed beyond the existing MySQL	0
OBJ-06	Complete audit trail	Sensitive actions recorded in the audit log	100%

In scope

- Time in/out capture, work schedules and holiday calendar
- Attendance tracking: late, undertime, overtime and absence
- Leave types, balances, requests and approval
- Payroll runs per pay period: earnings, deductions, statutory contributions and tax
- Payslip PDF generation and employee self-service viewing
- Audit trail, outbox push to the DMS, PMS MySQL database and private payslip storage

Out of scope

- Employee master records (EMS) and any student access
- Biometric hardware and drivers; only file import is considered [TBD]
- Bank crediting files and government remittance filings [TBD]

- External sharing of payslips, SSO and MFA (follow the DMS decision with the Enterprise Architect)

Requirements

Priorities follow MoSCoW and are proposed for client confirmation, as in the DMS document. Volumes and peak loads (employees, punches per day, payroll run size) are not known and must come from the client.

Functional requirements

ID	Module	Requirement	Priority
FR-01	Time in/out	Employee records time in and time out from the web; the server clock sets the time and every punch is stored as an append-only record	Must
FR-02	Time in/out	Missed or wrong punches are fixed through a correction request that a supervisor approves; the original punch stays	Must
FR-03	Attendance	Daily attendance is computed from punches, schedule and holidays: late, undertime, overtime, absence and rest day	Must
FR-04	Attendance	Employees, supervisors and the payroll officer view attendance per period; the payroll officer locks the period before a run	Must
FR-05	Leave	Leave types, yearly entitlement and balances kept as a ledger of credits and debits	Must
FR-06	Leave	Employee files a leave request; the supervisor approves or rejects; balance is checked on submit and debited on approval; employee can cancel before the start date	Must
FR-07	Payroll	Payroll run per pay period computes earnings, attendance deductions, statutory contributions, tax and net pay from locked inputs	Must
FR-08	Payroll	A run moves through Draft, Computed, Reviewed, Approved and Finalized; a finalized run cannot be edited, corrections go in a later run	Must

ID	Module	Requirement	Priority
FR-09	Payslips	Payslip PDF generated for each employee when a run is finalized; employee views and downloads only their own	Must
FR-10	Integration	Finalized payslips, payroll history events and leave decisions are sent to the DMS through the outbox	Must
FR-11	Accounts	Admin creates accounts, changes roles and activates or deactivates; no self-demotion or self-deactivation	Must
FR-12	Audit	Record actor, action and record for punch corrections, leave decisions, payroll runs, payslip access and role changes	Must
FR-13	Notifications	Email or in-app notice for leave decisions and payslip availability	Should
FR-14	Reports	Attendance summary, leave summary and payroll register exports	Should

Non-functional requirements

ID	Category	Requirement
NFR-01	Security	Session sign-in with active-account check, CSRF protection and sign-in throttling, same as the DMS
NFR-02	Integrity	Punches, finalized payroll lines and payslip files are never changed; SHA-256 checksum recorded for each payslip and re-verified on access
NFR-03	Idempotency	Resent events to the DMS return the original receipt; each event carries a stable event ID and an increasing revision per record
NFR-04	Accuracy	Money held as fixed-point decimals; rounding rule agreed with the client [TBD]; statutory rate tables carry effective dates
NFR-05	Limits	DMS API throttle 120 requests per minute per connector; payslip file up to 10 MB; history payload up to 256 KiB (limits inherited from the DMS)
NFR-06	Time	All times stored in UTC and shown in Asia/Manila; the browser clock is never trusted

ID	Category	Requirement
NFR-07	Availability	[TBD]; clock-in must stay usable at shift start, the usual daily peak
NFR-08	Recovery	MySQL and private payslip files restored together; RPO and RTO [to be agreed]
NFR-09	Privacy	Compliance with the Philippine Data Privacy Act of 2012 (RA 10173) [to confirm]; payroll data is Restricted
NFR-10	Accessibility	Keyboard navigation and readable text; formal WCAG 2.2 AA subject to client approval

Tier and component responsibilities

Only the Application tier talks to the Data tier, and the browser reaches the Application tier only through the Presentation tier. The DMS document counts a separate Integration tier; here the DMS-facing code is a module inside the Application tier so the design stays at three tiers (AD-01).

Tier	Contents
1. Presentation	Blade + Bootstrap 5.3 CSS + jQuery 3.7 + jQuery UI 1.14: clock page, attendance, leave, payroll console, payslip viewer, admin screens; session, CSRF and sign-in throttle
2. Application	Laravel 12 on PHP 8.2: controllers, domain services, payroll engine, audit service, scheduler, database-backed queue worker and the outbox publisher that calls the DMS /api/v1
3. Data	One PMS MySQL (InnoDB) database and private payslip file storage

Component	Responsibility	Traces to
Presentation views	Sign-in, role-aware navigation, clock page, attendance and leave screens, payroll console, payslip viewer	FR-01, FR-04, FR-06, FR-09

Component	Responsibility	Traces to
Session security	Active account, CSRF, sign-in throttle, department and role scoping	NFR-01, FR-11
TimeClockController + AttendanceService	Stamp server time, append punches, block duplicate punches, handle correction requests, compute daily attendance	FR-01 to FR-04
LeaveController + LeaveService	Leave types, ledger balances, requests, approval, cancellation, overlap check	FR-05, FR-06
PayrollController + PayrollEngine	Period lock, run states, earnings and deductions, statutory table lookup, adjustments, net pay	FR-07, FR-08, NFR-04
PayslipController + PayslipService	Render PDF, record SHA-256, store privately, serve only to the owner	FR-09, NFR-02
AccountController	Admin-only account creation, roles and activation	FR-11
AuditService	Write audit entries for sensitive actions	FR-12
OutboxPublisher	Send outbox events and payslip files to the DMS, retry with backoff and jitter	FR-10, NFR-03
Scheduler + queue worker	Nightly attendance computation, payroll batch, payslip generation and outbox retry, on the MySQL queue driver	OBJ-05
PMS MySQL	Transactions, unique constraints, revision compare-and-swap	NFR-02, NFR-03
Private payslip storage	UUID paths outside public/, checksum on access	NFR-02

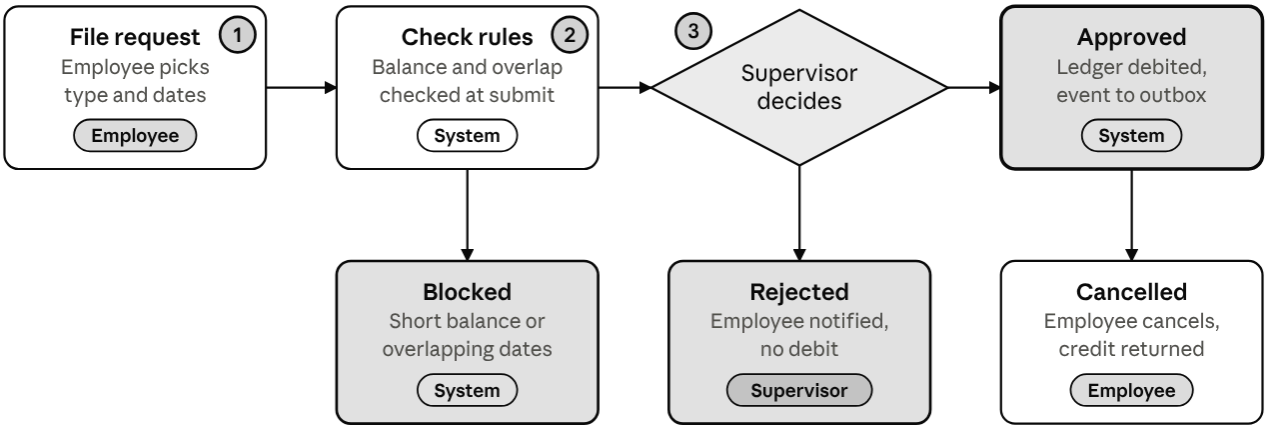
Core workflows

Time in/out and attendance

1. The employee presses Time in or Time out on the clock page and the server stamps the time.
2. A duplicate-punch guard rejects a second punch of the same kind within a short window [length to agree].
3. A nightly job pairs punches with the schedule and holidays and writes the attendance day.
4. A missed or wrong punch goes to the supervisor as a correction request; approval recomputes that day.
5. At period lock, attendance days freeze and feed the payroll run.

Leave request

Leave is debited only after the supervisor approves

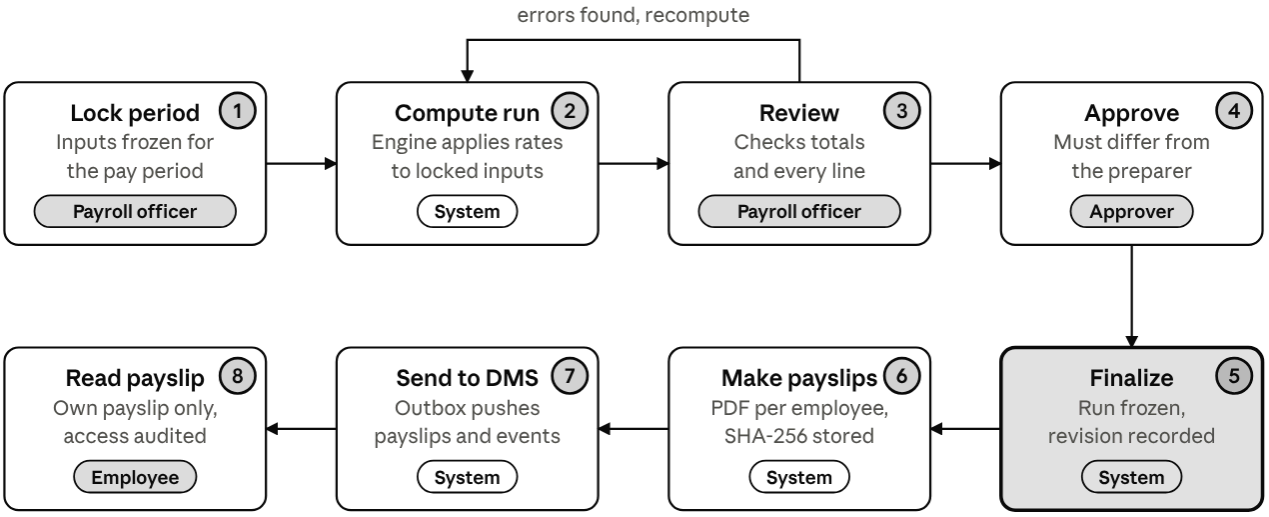


leave request path · 1 decision, 3 outcomes

An approved leave debits the ledger and marks the days as leave in attendance, which the next payroll run reads.

Payroll run

A payroll run cannot change once it is finalized



payroll run · 8 steps, 1 loop

The payroll officer locks the period and the engine computes the run. A different person approves it, and finalizing freezes it, so a mistake needs an adjustment in the next run or a new revision.

Data architecture

The PMS is the system of record for punches, attendance, leave, payroll runs and payslips. The EMS stays the system of record for employees; the PMS keeps only a payroll profile keyed by the EMS employee ID. Classes are proposed and need client sign-off.

Entity	Key attributes	Change rule	Class
User account	id, email, password hash, role, active, department, campus	Admin edits	Restricted
Employee payroll profile	employee_ref (EMS ID), pay group, basic rate, schedule, supervisor, tax and contribution settings, effective_from	New row per change, effective-dated	Restricted
Work schedule, holiday	shift times, rest days, holiday date and type	Effective-dated edits by HR	Internal
Time punch	UUID, employee_ref, in/out, server time (UTC), source	Append-only, never updated	Confidential

Entity	Key attributes	Change rule	Class
Punch correction request	employee, date, proposed in/out, reason, status, decided_by	Status changes; the original punch stays	Confidential
Attendance day	employee, date, first in, last out, late, undertime, overtime, status, revision	Recomputed until the period is locked, then frozen	Confidential
Leave type, ledger entry	type, paid flag, yearly credit; employee, type, delta, reason, request id	Ledger append-only	Confidential
Leave request	UUID, employee, type, dates, days, status, decided_by, revision	Status changes with revision	Confidential
Pay period	start, end, pay date, status (Open, Locked)	Lock is one-way	Internal
Payroll run	UUID, period, status, revision, totals, approved_by	State machine; frozen at Finalized	Restricted
Payroll line	run, employee, item code, kind (earning, deduction, contribution, tax), amount	Immutable once the run is Finalized	Restricted
Statutory rate table	type, bracket, rate, effective_from, effective_to	New version per change, never overwritten	Internal
Payslip	UUID, run, employee, version, storage_key, SHA-256, DMS document ref	Immutable; a correction is a new version	Restricted
Audit entry	actor, action, record, time	Append-only	Confidential
Outbox event	event ID, record key, revision, payload, status, attempts, DMS receipt ID	Append-only apart from status and attempts	Confidential

Rules that keep the records trustworthy

- A punch is never edited or deleted; a correction is a separate record that a supervisor approves, and the attendance day is recomputed from both.

- Once a pay period is locked, its attendance days and approved leave for that period stop changing.
- A Finalized run is read-only. A mistake is fixed by an adjustment line in the next run, or by a new run revision with a stated reason [client to choose].
- Leave reasons that reveal health details are visible only to the approver and HR.
- Retention periods, data residency and subject access handling are [to be agreed with the client], and a Data Protection Impact Assessment is completed before go-live.

Integration with the DMS

In the DMS document the PMS is one of the four source systems. It reaches the DMS only through /api/v1 with a bearer token bound to one source, school and campus, and never reads another ERP database. The outbox keeps events safe while the DMS is unreachable.

ID	Interface	Direction	Pattern	Frequency	Data
PINT-01	History events	PMS → DMS	REST push, idempotent, from the outbox	Per record change	Leave decisions, payroll run state changes, period locks
PINT-02	Payslip documents	PMS → DMS	REST push, idempotent, from the outbox	On run finalization	Payslip PDF, SHA-256, run and employee reference
PINT-03	Employee read-back	DMS → PMS (GET)	REST, cursor-based	Scheduled and on demand	EMS employee snapshots [schema to agree with the EMS team]
PINT-04	Reconciliation	DMS → PMS (GET)	REST, cursor-based	On demand	Received history and events for the PMS source

Delivery rules

- The PMS writes its own change and an outbox row in one MySQL transaction, so a change is never made without its event.
- Every event has a stable event ID and the record's increasing revision; the publisher sends it with backoff and jitter until the DMS answers.
- A new event returns 201. An identical replay returns 200 with the original receipt. The same event with a different hash, or a different event at the same revision, returns 409 and is held for operator review [who is alerted: to be defined].
- An out-of-order older revision is stored by the DMS without changing the latest snapshot, so the publisher never has to wait for earlier events.

- Throttle is 120 requests per minute; the publisher sends in small batches and pauses on a 429.
- Payslips in the DMS are visible to payroll officers of the PMS source and to admins under the DMS role matrix. Employees read their own payslips in the PMS, since DMS employee roles have no source-document access.
- A long DMS outage only delays delivery: events stay in the outbox until sent.

Security and roles

Staff sign in with school accounts in a session, as in the DMS. Payroll data is Restricted, so access is limited by role, school and campus on every query, and the person who computes a payroll run cannot approve it.

Role	Punches and attendance	Leave	Payroll runs	Payslips	Accounts and audit
Employee	Own: clock, view, request correction	Own requests and balance	None	Own only	None
Supervisor	Own and team: approve corrections	Own and team: approve or reject	None	Own only	None
HR officer	Read all in scope; manage schedules and holidays	Manage types and balances; read requests	None	None	None
Payroll officer	Read all in scope; lock the period	Read approved leave	Create, compute, review	Generate; read all in scope	None
Payroll approver [role holder to confirm]	Read summary	None	Approve and finalize	Read all in scope	None
Admin	None	None	None	None	Manage accounts; read audit

Controls

- Identity: session sign-in with active-account check and throttling; no self-demotion or self-deactivation; SSO and MFA are not in the current design, though MFA for payroll roles is recommended [decide with the Enterprise Architect].
- Time integrity: the server sets punch time, a duplicate-punch guard blocks double taps, and punches cannot be edited. Limiting the clock page to campus networks is an option [client to decide].

- Payslips: private storage outside public/, UUID paths, SHA-256 re-checked on every access, owner-only reads, each read and download audited.
- Application: CSRF protection, API throttle, hashed connector token, and school, campus and role scoping inside every query.
- Transport: only the public/ directory is served, over HTTPS.
- Not yet decided: encryption at rest, central monitoring and penetration testing.

Infrastructure and deployment

The hosting model is still open with the Enterprise Architect, so this section lists what the PMS needs wherever it runs. It matches the DMS platform: no managed database, queue, search or object storage.

- Compute: one PHP 8.2 / Laravel 12 application; only public/ is served; one scheduler entry and one queue worker process.
- Data: one MySQL (InnoDB) database and a private folder for payslip files; the queue uses the MySQL driver.
- Tenancy: school and campus scoping inside one deployment.
- Environments: Dev and Test on synthetic data, UAT on an anonymised copy, Production on real data; hosting for each [pending Enterprise Architect].
- Backups: MySQL and private payslip files are backed up and restored together, because the database holds each payslip's storage key and checksum; restore frequency [to be agreed].
- Monitoring: [to be defined]; watch the outbox for events stuck in Pending or Failed and runs stuck in Computed.
- Reachability: the PMS must reach the DMS /api/v1; the outbox covers outages.
- Capacity: payroll run size and clock-in peak at shift start are unknown and need load testing.

Risks, assumptions and open questions

Ratings are proposed for client confirmation.

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-01	Statutory rates or payroll rules wrong or out of date	Medium	Very high	Effective-dated rate tables; parallel run checked against manual computation before go-live	Payroll officer

ID	Risk	Likelihood	Impact	Mitigation	Owner
R-02	Disputed or tampered time records	Medium	High	Append-only punches, approved corrections, audit entries	PMS team
R-03	Clock-in peak at shift start slows the app	Medium	Medium	Light punch endpoint; load test	PMS team
R-04	EMS adapter late, so employee data is missing	Medium	High	Local payroll profile with manual import as fallback	EMS team
R-05	Payroll data exposed	Low	Very high	Role and campus scoping, private storage, DPIA, audited access	Client / DPO [TBD]
R-06	MySQL and payslip files restored out of step	Medium	High	Joint backup and tested restore	PMS Solutions Architect
R-07	DMS outage or conflicts delay payslip delivery	Medium	Medium	Outbox with retries; 409 review queue	PMS team

Assumptions

- The EMS supplies a stable employee ID, department and supervisor for each employee.
- Users are school staff on school-managed devices; students have no access.
- The PMS builds its own source adapter and keeps revisions increasing per record.
- The client and the Enterprise Architect decide hosting, budget and data residency.

Open questions for the client

- Pay frequency and cutoff dates, and employee types (plantilla, contractual, part-time teaching on hourly rates)
- Which statutory contributions and tax rules apply (for example GSIS or SSS, PhilHealth, Pag-IBIG, withholding tax)
- Overtime, holiday and night-differential rules, and the leave policy (types, accrual, carry-over)
- How long after a missed punch a correction may be filed
- Whether bank crediting files or biometric import are needed at go-live

Architecture decisions log

ID	Decision	Options considered	Rationale	Date
AD-01	One Laravel application in three tiers (Presentation, Application, Data); DMS-facing code sits inside the Application tier	Four tiers as in the DMS; microservices	Meets the three-tier requirement and keeps one deployable unit; the DMS boundary stays a single module	2026-10-16
AD-02	Punches are append-only; corrections are separate approved records	Editable punch rows	Protects the evidence behind pay and supports OBJ-01	2026-10-16
AD-03	Payroll runs follow a one-way state machine and freeze at Finalized	Editable runs with a change log	Keeps results repeatable and defensible (OBJ-02)	2026-10-16
AD-04	Payslip PDFs are rendered on the server, stored privately and checksummed with SHA-256	Render in the browser; managed document service	Same integrity model as DMS versions, and no vendor cost	2026-10-16
AD-05	Outbox pattern for every DMS call	Direct call inside the request	A change and its event commit together; DMS outages cause delay, not loss (OBJ-04)	2026-10-16
AD-06	MySQL queue driver and Laravel scheduler for batch work	Redis or a managed queue	No extra engine, in line with OBJ-05	2026-10-16